

DualCheck: Certificates and Skolem Strategies for Expansion-Based QBF Solving

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Abstract. Keywords: Quantified Boolean Formulas · Proof Certification · Expansion-Based Solving · Strategy Extraction.

1 Preliminaries and Background

1.1 Quantified Boolean Formulas

A *quantified Boolean formula* (QBF) in *prenex normal form* (PNF) is an expression of the form $\Phi = \Pi.\psi$, where Π is the *quantifier prefix* and ψ is a quantifier-free propositional formula called the *matrix*. The prefix has the form $\Pi = Q_1v_1 \dots Q_nv_n$, where each $Q_i \in \{\forall, \exists\}$ is a quantifier and $V_\Phi = \{v_1, \dots, v_n\}$ is a set of pairwise distinct Boolean variables. The prefix defines a strict linear order $<_\Pi$ on V_Φ . The set V_Φ is partitioned into existential variables $E_\Phi = \{v_i \mid Q_i = \exists\}$ and universal variables $U_\Phi = \{v_i \mid Q_i = \forall\}$. The *dependency domain* of a variable $v \in V_\Phi$ under prefix Π is the set of strictly preceding variables:

$$D_\Pi(v) = \{w \in V_\Phi \mid w <_\Pi v\}. \quad (1)$$

We define $\text{vars}(\cdot)$ as the function that returns the set of variables in a propositional formula, and we assume all formulas are closed, meaning all variables occurring in the matrix appear in the prefix.

Literals, Clauses, and Cubes. A *literal* ℓ is either a variable $v \in V_\Phi$ (positive literal) or its negation $\neg v$ (negative literal). The underlying variable of ℓ is denoted by $\text{vars}(\ell) = v$, and its complement is $\bar{\ell}$ (where $\bar{v} = \neg v$ and $\overline{\neg v} = v$). A *clause* is a finite disjunction of literals, and a *cube* is a finite conjunction of literals (both often treated as sets of literals). A propositional formula is in *conjunctive normal form* (CNF) if it is a conjunction of clauses, and in *disjunctive normal form* (DNF) if it is a disjunction of cubes.

When the matrix ϕ of $\Phi = \Pi.\phi$ is in CNF, Φ is a *prenex conjunctive normal form* (PCNF) QBF. For each clause $C \in \phi$, we partition its literals into existential and universal subclauses:

$$C = C_E \vee C_U, \quad \text{vars}(C_E) \subseteq E_\Phi, \quad \text{vars}(C_U) \subseteq U_\Phi. \quad (2)$$

Dually, when the matrix is in DNF, the formula is in *prenex disjunctive normal form* (PDNF).

Assignments and Truth Semantics. An *assignment* τ on a set of variables $W \subseteq V_\Phi$ is a function $\tau : W \rightarrow \{\perp, \top\}$. For any subset $W' \subseteq W$, the restriction of τ to W' is denoted by $\tau|_{W'}$. We write $\tau|_{<_\Pi v} = \tau|_{D_\Pi(v) \cap W}$ for the restriction of τ to variables preceding v under $<_\Pi$. A total assignment $\tau : V_\Phi \rightarrow \{\perp, \top\}$ satisfies a clause C (written $\tau \models C$) if $\tau(\ell) = \top$ for at least one literal $\ell \in C$.

An assignment $\alpha : U_\Phi \rightarrow \{\perp, \top\}$ to all universal variables is called a *universal assignment*, and an assignment $\sigma : E_\Phi \rightarrow \{\perp, \top\}$ to all existential variables is called an *existential assignment*. The truth of a closed QBF is defined inductively on the prefix length: a formula with an empty prefix is a variable-free sentence, which is true if it evaluates to $\top$ and false if it evaluates to $\perp$. For a formula $Qv \Pi.\psi$, it is true if:

- $Q = \exists$ and $\Pi.\psi[v/\top] \vee \Pi.\psi[v/\perp]$ is *true*;
- $Q = \forall$ and $\Pi.\psi[v/\top] \wedge \Pi.\psi[v/\perp]$ is *true*.

Formula Negation. For a quantifier $Q \in \{\forall, \exists\}$, let $\overline{Q}$ denote its dual quantifier ($\overline{\forall} = \exists$ and $\overline{\exists} = \forall$). The *negation* of a PCNF formula $\Phi = \Pi.\phi$ with $\phi = \{C_1, \dots, C_m\}$ is the PDNF formula:

$$\neg\Phi = \overline{\Pi}.\neg\phi, \quad \overline{\Pi} = \overline{Q}_1 v_1 \dots \overline{Q}_n v_n, \quad \neg\phi \equiv \bigvee_{C \in \phi} \neg C. \quad (3)$$

Because each $\neg C = \bigwedge_{\ell \in C} \bar{\ell}$ is a cube, $\neg\phi$ is a DNF matrix. Under negation, quantifier roles invert: $U_{\neg\Phi} = E_\Phi$ and $E_{\neg\Phi} = U_\Phi$, while the prefix order $<_{\overline{\Pi}} = <_\Pi$ is preserved. Semantically, $\neg\Phi$ is true if and only if Φ is false.

1.2 QBF Strategies

Definition 1 (Strategies). Let $\Phi = \Pi.\phi$ be a PCNF QBF. An admissible strategy for a quantified variable set $X \in \{E_\Phi, U_\Phi\}$ is a tuple of Boolean formulas $f = (f_x)_{x \in X}$ such that each f_x depends only on preceding variables in the prefix:

$$\text{vars}(f_x) \subseteq D_\Pi(x). \quad (4)$$

We write $\phi[f]$ for the propositional formula obtained by sequentially substituting f_x for $x \in X$ in reverse prefix order.

- An admissible strategy $s = (s_e)_{e \in E_\Phi}$ is a winning Skolem strategy for Φ if $\phi[s] \equiv \top$.
- An admissible strategy $h = (h_u)_{u \in U_\Phi}$ is a winning Herbrand strategy for Φ if $\phi[h] \equiv \perp$.

A closed QBF Φ is true if and only if it admits a winning Skolem strategy, and false if and only if it admits a winning Herbrand strategy [1].

Remark 1 (Strategy Duality under Negation). Because quantifier types invert under negation ($U_{\neg\Phi} = E_\Phi$) while the prefix order is preserved ($D_{\overline{\Pi}} = D_\Pi$), any admissible Herbrand strategy h for $\neg\Phi$ satisfies $\text{vars}(h_e) \subseteq D_\Pi(e)$ for each

$e \in E_\Phi$. Since substitution commutes with negation, $(\neg\phi)[h] \equiv \neg(\phi[h])$, which yields:

$$(\neg\phi)[h] = \perp \quad \text{if and only if} \quad \phi[h] = \top. \quad (5)$$

Therefore, h is a winning Herbrand strategy for $\neg\Phi$ if and only if it is a winning Skolem strategy for Φ .

1.3 Expansion Trees and Expansion-Based Solving

Expansion-based solving eliminates quantifiers by systematically unrolling the prefix into a propositional formula over annotated variable copies [2, 4, 5]. Rather than expanding all quantifiers simultaneously, solvers expand variables of one quantifier kind at a time along assignment branches.

Given a universal assignment branch $\alpha : U_\Phi \rightarrow \{\perp, \top\}$, each existential variable $x \in E_\Phi$ is replicated as an annotated propositional variable x^ω , where $\omega = \alpha|_{<_{\Pi} x}$ records the values of universal variables preceding x . For an existential literal ℓ with $x = \text{vars}(\ell)$, the annotated literal ℓ^ω denotes x^ω if $\ell = x$, and $\neg x^\omega$ if $\ell = \neg x$.

Instantiating $\Phi = \Pi.\phi$ along a universal branch α yields a propositional CNF formula $\phi^\alpha = \bigwedge_{C \in \phi} C^\alpha$ over annotated existential variables, where:

$$C^\alpha = \begin{cases} \top & \text{if } \alpha \models C_U, \\ \{\ell^{\alpha|_{<_{\Pi} \text{vars}(\ell)}} \mid \ell \in C_E\} & \text{if } \alpha \not\models C_U. \end{cases} \quad (6)$$

An *expansion tree* represents a collection of branches. The complete universal expansion tree encompasses all $2^{|U_\Phi|}$ universal assignments: $\exp(\Phi) = \bigwedge_{\alpha: U_\Phi \rightarrow \{\perp, \top\}} \phi^\alpha$. A QBF Φ is false if and only if there exists a finite set of universal assignment branches $A \subseteq \{U_\Phi \rightarrow \{\perp, \top\}\}$ such that the partial expansion $\bigwedge_{\alpha \in A} \phi^\alpha$ is propositionally unsatisfiable [2, 4].

Dually, evaluating Φ along an existential assignment branch $\sigma : E_\Phi \rightarrow \{\perp, \top\}$ yields a propositional formula ϕ^σ over universal variables. A QBF Φ is true if and only if there exists a set of existential branches $S \subseteq \{E_\Phi \rightarrow \{\perp, \top\}\}$ such that the disjunction $\bigvee_{\sigma \in S} \phi^\sigma$ is valid, or equivalently, the conjunction of negated instantiations is unsatisfiable:

$$\bigwedge_{\sigma \in S} \neg\phi^\sigma \equiv \perp. \quad (7)$$

Modern expansion solvers (e.g., RAReQS [4], Ijtihad [2]) solve QBFs using two propositional SAT instances in parallel within a Counterexample Guided Abstraction Refinement (CEGAR) loop:

- The *primal instance* accumulates instantiations $\bigwedge_{\alpha \in A} \phi^\alpha$ under universal branches $\alpha : U_\Phi \rightarrow \{\perp, \top\}$. A propositional refutation of this instance proves that Φ is false.
- The *dual instance* accumulates negated instantiations $\bigwedge_{\sigma \in S} \neg\phi^\sigma$ under existential branches $\sigma : E_\Phi \rightarrow \{\perp, \top\}$. A propositional refutation of this instance proves that Φ is true.

1.4 Expansion-Based Proof Systems

Expansion proof systems combine quantifier expansion rules with an underlying propositional proof system $\mathcal{P}$ [5, 8]. The system $\forall\text{-Exp}+\mathcal{P}$ over a QBF $\Psi = \Pi.\psi$ is defined by:

1. **Axiom Rule (Universal Expansion):** Given a universal assignment branch $\alpha : U_\Psi \rightarrow \{\perp, \top\}$, instantiating the universal variables of Ψ under α yields the propositional formula $\psi^\alpha = \psi[\alpha]$. When Ψ is in PCNF, for each clause $C = C_E \vee C_U \in \psi$ with $\alpha \not\models C_U$, the rule introduces the annotated clause:

$$\frac{}{\{\ell^\alpha |_{<\Pi \text{ vars}(\ell)} \mid \ell \in C_E\}} \quad (\alpha \not\models C_U). \quad (8)$$

2. **Propositional Refutation in $\mathcal{P}$:** A refutation of the derived set of annotated clauses in $\mathcal{P}$, concluding with the empty clause $\perp$.

When $\mathcal{P}$ is propositional resolution (Res), this instantiates $\forall\text{-Exp}+\text{Res}$ [5], where resolution is performed on annotated existential variables with identical annotations:

$$\frac{C_1 \vee x^\omega \quad C_2 \vee \neg x^\omega}{C_1 \vee C_2} \quad (x \in E_\Psi, \omega = \alpha|_{<\Pi x}). \quad (9)$$

The $\forall\text{-Exp}+\text{Res}$ calculus is sound and refutationally complete for false PCNF QBFs [5]. Replacing $\mathcal{P}$ with clausal checking systems such as RUP or RAT yields $\forall\text{-Exp}+\text{RUP}$ and $\forall\text{-Exp}+\text{RAT}$ [8].

1.5 Expansion Certification for False QBFs

To verify solver results independently of solver heuristics, certification frameworks decouple verification into a two-phase pipeline [3, 8].

Certificate Contents. An expansion certificate for a false QBF $\Phi = \Pi.\phi$ consists of:

1. An *annotation map* defining the provenance of each propositional variable v appearing in the proof via:
 - A *variable mapping* $\varepsilon(v) \in E_\Phi$, identifying the original existential variable represented by v . We extend ε homomorphically to literals by $\varepsilon(\neg v) = \neg\varepsilon(v)$, and to clauses by $\varepsilon(C) = \{\varepsilon(\ell) \mid \ell \in C\}$.
 - An *assignment mapping* $\text{ann}(v)$, recording the universal branch prefix under which v was instantiated.
2. An *axiom clause set*, containing the annotated clauses generated by universal expansion.
3. An *origin map* orig , associating each axiom clause with the index of the matrix clause of Φ from which it was instantiated.
4. A *propositional refutation* in proof system $\mathcal{P}$ deriving $\perp$ from the axiom clause set.

Two-Phase Verification Pipeline. Verification proceeds in two phases:

1. **Phase 1 (Expansion Check):** An independent polynomial-time algorithm validates that every clause C in the axiom clause set is a legal universal expansion axiom of Φ . For each clause C , the checker retrieves $D = \phi[\text{orig}(C)]$ and verifies:
 - *Existential literal correspondence:* $\varepsilon(C) = D_E$.
 - *Universal literal falsification:* for $\tau = \bigcup_{\ell \in C} \text{ann}(\text{vars}(\ell))$, the checker confirms:

$$\forall \ell \in D_U, \quad \neg \ell \in \tau. \quad (10)$$

2. **Phase 2 (Propositional Refutation Check):** An external certifying checker (e.g., `drat-trim`) validates that $\mathcal{P}$ derives $\perp$ soundly from the verified axiom clauses.

2 Certifying Expansion Refutations (Proof Checking)

In this section, we present the proof-checking foundations of DualCheck. We first analyze the structural obstacle preventing direct certification in the true case, and formalize the solver’s dual instance as a well-defined QBF denoted $\text{PG}(\neg\Phi)$. We then prove the Axiom Correspondence Lemma, establishing that every clause emitted by an expansion solver into its dual instance is an exact $\forall\text{-Exp}+\text{Res}$ axiom of $\text{PG}(\neg\Phi)$. Finally, we introduce the unified DualCheck proof format (`.dc`) and the two-phase verification algorithms that validate both true and false solver outcomes.

2.1 The Clausal Barrier in True QBF Solving

As established in Section 1.3, modern expansion solvers decide whether a QBF $\Phi = \Pi.\phi$ is true by refuting its dual instance $\bigwedge_{\sigma \in S} \neg\phi^\sigma$, accumulated over candidate existential witnesses $\sigma : E_\Phi \rightarrow \{\perp, \top\}$. Recall that $U_{\neg\Phi} = E_\Phi$; thus, an existential candidate assignment σ is identically a universal assignment to the universal variables of the negation $\neg\Phi$.

Under a candidate witness $\sigma : E_\Phi \rightarrow \{\perp, \top\}$, each clause $C_i = C_{i,E} \vee C_{i,U} \in \phi$ is evaluated on its existential literals:

- If $\sigma \models C_{i,E}$, the clause is satisfied by σ ($C_i^\sigma = \top$), and provides no constraint.
- If $\sigma \not\models C_{i,E}$, the clause survives as its universal subclause $C_{i,U}$.

Let $\phi^\sigma = \{C_i \in \phi \mid \sigma \not\models C_{i,E}\}$ denote the set of matrix clauses unsatisfied by σ . The formula added to the dual instance on branch σ is the negation of their conjunction:

$$\neg\phi^\sigma = \bigvee_{C_i \in \phi^\sigma} \neg C_{i,U}^\sigma = \bigvee_{C_i \in \phi^\sigma} \bigwedge_{\ell \in C_{i,U}} \neg \ell^\sigma. \quad (11)$$

Because each term $\bigwedge_{\ell \in C_{i,U}} \neg \ell^\sigma$ is a conjunction of literals, $\neg\phi^\sigma$ is a disjunction of cubes—a DNF formula. Converting $\neg\phi^\sigma$ into CNF via distributivity would incur an exponential explosion in clause count.

To preserve polynomial size, expansion solvers apply a structure-preserving Plaisted–Greenbaum (PG) transformation [6]. Each universal subclause $C_{i,U}$ is associated with a fresh auxiliary variable t_i , so that the negated clause—that is, the cube $\neg C_{i,U} = \bigwedge_{\ell \in C_{i,U}} \neg \ell^\sigma$ —is represented by the auxiliary literal $\neg t_i^\sigma$. The negated conjunction $\neg \phi^\sigma$ is then represented clausally by the reference clause:

$$R^\sigma = \bigvee_{C_i \in \phi^\sigma} \neg t_i^\sigma, \quad (12)$$

coupled with the definition clauses $(t_i \vee \neg \ell)$ for each literal $\ell \in C_{i,U}$. Because $(t_i \vee \neg \ell)$ is equivalent to $\neg t_i \rightarrow \neg \ell$, these definition clauses encode the implication:

$$\neg t_i \rightarrow \bigwedge_{\ell \in C_{i,U}} \neg \ell, \quad (13)$$

asserting that the auxiliary literal $\neg t_i$ in the reference clause implies the negated clause, i.e., the cube $\neg C_{i,U}$.

Consequently, the dual SAT solver refutes instantiations not of the input formula Φ , but of an encoded formula containing auxiliary variables t_i . This transformation breaks the direct clause-to-clause correspondence that false-case certification frameworks rely upon: the reference clauses R^σ and definition clauses do not exist in the original matrix ϕ .

2.2 The Encoded Negation $\text{PG}(\neg\Phi)$ and Axiom Correspondence

To place dual refutations on a rigorous proof-theoretic foundation, we formalize the solver’s clausal encoding at the QBF level. Refuting the dual instance witnesses that Φ is true, which is semantically equivalent to refuting the negation $\neg\Phi = \overline{\Pi}.\neg\phi$. The formula $\text{PG}(\neg\Phi)$ lifts the solver’s auxiliary variables and definition clauses into a formal QBF.

Definition 2 (Plaisted–Greenbaum Encoding $\text{PG}(\neg\Phi)$). *Let $\Phi = \Pi.\phi$ be a PCNF QBF with $\phi = \{C_1, \dots, C_m\}$. For each clause $C_i \in \phi$, let t_i be an auxiliary variable associated with C_i such that $\neg t_i$ represents the negated universal subclause (cube) $\neg C_{i,U} = \bigwedge_{\ell \in C_{i,U}} \neg \ell$, and let $T = \{t_1, \dots, t_m\}$. The QBF $\text{PG}(\neg\Phi) = \Pi'.\phi'$ is defined by:*

- **Prefix:** Π' is obtained from the dual prefix $\overline{\Pi}$ by inserting each auxiliary variable t_i existentially immediately after $v_i^* = \max_{<\Pi} \text{vars}(C_{i,U})$, so that

$$U_{\text{PG}(\neg\Phi)} = U_{\neg\Phi} = E_\Phi, \quad E_{\text{PG}(\neg\Phi)} = E_{\neg\Phi} \cup T = U_\Phi \cup T. \quad (14)$$

- **Matrix:** ϕ' consists of the definition clauses and the reference formula:
 1. Definition clauses: for each clause $C_i \in \phi$ and each literal $\ell \in C_{i,U}$, the binary clause:

$$(t_i \vee \neg \ell), \quad (15)$$

encoding the implication $\neg t_i \rightarrow \neg \ell$, and collectively $\neg t_i \rightarrow \bigwedge_{\ell \in C_{i,U}} \neg \ell$ (i.e., $\neg t_i \rightarrow \neg C_{i,U}$).

2. Reference formula: *the formula*

$$\bigvee_{C_i \in \phi} (\neg C_{i,E} \wedge \neg t_i). \quad (16)$$

Quantifying each t_i immediately after v_i^* ensures that every variable in the definition of t_i precedes t_i in the prefix Π' :

$$\forall \ell \in C_{i,U}, \quad \text{vars}(\ell) \leq_{\Pi'} v_i^* <_{\Pi'} t_i. \quad (17)$$

Because each auxiliary variable t_i is existential and quantified after all variables in its defining subclause $C_{i,U}$, setting $t_i \equiv \bigvee_{\ell \in C_{i,U}} \ell$ is prefix-admissible, reduces each definition clause $(t_i \vee \neg \ell)$ to a tautology, and transforms the reference formula $\bigvee_{C_i \in \phi} (\neg C_{i,E} \wedge \neg t_i)$ into $\bigvee_{C_i \in \phi} \neg C_i \equiv \neg \phi$. Hence, $\text{PG}(\neg \Phi)$ is an equisatisfiable definitional encoding of $\neg \Phi$, false if and only if $\neg \Phi$ is false.

When the dual solver explores a candidate assignment $\sigma : U_{\neg \Phi} \rightarrow \{\perp, \top\}$, it accumulates definition clauses and reference clauses over annotated existential variables. We now prove that all such clauses are valid universal expansion axioms of $\text{PG}(\neg \Phi)$.

Lemma 1 (Axiom Correspondence). *Every clause introduced into the dual instance by an expansion solver on branch σ is a valid universal expansion axiom of $\text{PG}(\neg \Phi)$ under the universal assignment $\sigma : U_{\neg \Phi} \rightarrow \{\perp, \top\}$.*

Proof. We analyze the two classes of clauses generated on branch σ :

1. **Definition clauses.** For each clause $C_i \in \phi$ and literal $\ell \in C_{i,U}$, the binary clause $(t_i \vee \neg \ell)$ is an explicit clause of the matrix ϕ' of $\text{PG}(\neg \Phi)$. Both $t_i \in T$ and $\text{vars}(\ell) \in U_{\Phi} = E_{\neg \Phi}$ are existential variables in $\text{PG}(\neg \Phi)$; neither is assigned by the universal assignment $\sigma : U_{\neg \Phi} \rightarrow \{\perp, \top\}$. Under universal assignment σ , the $\forall$ -Exp+Res axiom rule simply annotates each literal with the restriction of σ to preceding universal variables in Π' , yielding the annotated clause:

$$(t_i^{\sigma|<_{\Pi'} t_i} \vee \neg \ell^{\sigma|<_{\Pi'} \text{vars}(\ell)}). \quad (18)$$

Hence, every definition clause emitted by the solver is a valid $\forall$ -Exp+Res axiom of $\text{PG}(\neg \Phi)$.

2. **Reference clauses.** On branch σ , the solver emits the reference clause:

$$R^\sigma = \bigvee_{C_i \in \phi^\sigma} \neg t_i^{\sigma|<_{\Pi'} t_i}, \quad (19)$$

where $\phi^\sigma = \{C_i \in \phi \mid \sigma \not\models C_{i,E}\}$. In $\text{PG}(\neg \Phi)$, the reference formula is $\bigvee_{C_i \in \phi} (\neg C_{i,E} \wedge \neg t_i)$. Because $\text{vars}(C_{i,E}) \subseteq E_\Phi = U_{\neg \Phi}$, every literal in $C_{i,E}$ is universal in $\text{PG}(\neg \Phi)$. Under the full universal assignment $\sigma : U_{\neg \Phi} \rightarrow \{\perp, \top\}$, each term $(\neg C_{i,E} \wedge \neg t_i)$ is evaluated as follows:

- If $\sigma \models C_{i,E}$, then $\neg C_{i,E}$ evaluates to $\perp$, and the term evaluates to $\perp \wedge \neg t_i \equiv \perp$.

- If $\sigma \not\models C_{i,E}$ (i.e., $C_i \in \phi^\sigma$), then $\neg C_{i,E}$ evaluates to $\top$, leaving the unassigned existential literal $\neg t_i^{\sigma|_{<\Pi'^{t_i}}}$.

The disjunction of surviving terms yields precisely the reference clause R^σ . Therefore, under universal assignment σ , the reference formula simplifies under Boolean constant evaluation directly to the clause R^σ .

Thus, all initial clauses refuted in the dual instance are clauses of the instantiated matrix $\phi'[\sigma]$, confirming that they are valid universal expansion axioms of $\text{PG}(\neg\Phi)$. $\square$

Theorem 1 (Refutation Certification). *Let π be a propositional resolution refutation of the dual instance produced by an expansion solver. Then π is an $\forall\text{-Exp+Res}$ refutation of $\text{PG}(\neg\Phi)$, soundly certifying that $\text{PG}(\neg\Phi)$ is false and that Φ is true.*

Proof. By Lemma 1, every initial clause of π is a valid $\forall\text{-Exp+Res}$ axiom of $\text{PG}(\neg\Phi)$. Every resolution step in π operates on annotated existential literals with identical prefixes. Because π derives the empty clause $\perp$, π is an $\forall\text{-Exp+Res}$ refutation of $\text{PG}(\neg\Phi)$. By soundness of $\forall\text{-Exp+Res}$, $\text{PG}(\neg\Phi)$ is false. By Definition 2, $\text{PG}(\neg\Phi)$ is an equisatisfiable definitional encoding of $\neg\Phi$. Therefore, $\neg\Phi$ is false, which by QBF duality (Remark 1) soundly certifies that Φ is true. $\square$

2.3 The DualCheck Proof Format

To enable independent verification, DualCheck defines a unified proof format (`.dc`) to support both true and false QBFs. The format uses standard ASCII representation, organized into distinct sections:

Status and Problem Header. Every proof begins with a status line declaring whether the certified result is false or true:

`s 0` (false QBF) or `s 1` (true QBF).

This is immediately followed by the dimension header:

`p dc` `<num_prop_vars>` `<clause_count>`

specifying the dimensions of the proof:

- `<num_prop_vars>`: the highest propositional variable index used anywhere in the proof (following the standard DIMACS upper-bound convention);
- `<clause_count>`: the number of initial propositional clauses in the expansion formula—specifically, the count of axiom clauses (**a**) for false QBFs, or the combined count of definition (**h**) and reference (**n**) clauses for true QBFs, excluding subsequent derived resolution clauses (**d**).

Expansion Lines. Lines starting with **x** state the variables introduced in the certificate, what original QBF variables they represent, and the assignment under which they were instantiated:

$$\mathbf{x} \ v_1 \dots v_k \ 0 \ q_1 \dots q_k \ 0 \ l_1 \dots l_p \ 0$$

where $v_1 \dots v_k$ are the variables in the certificate, $q_1 \dots q_k$ are the original variables they represent, and $l_1 \dots l_p$ is the assignment under which they were instantiated. Variables appearing in clauses that are not declared in any **x** line are auxiliary definition variables T .

Expansion Clauses. Initial clauses are tagged by single-character identifiers and store all required semantic provenance inline:

- **Axiom clauses** (**a**, false QBFs):

$$\mathbf{a} \ id \ l_1 \dots l_k \ 0 \ orig \ 0$$

storing the sequential clause identifier id , the instantiated literals $l_1 \dots l_k$, and the 1-based index $orig$ of the originating matrix clause $D \in \phi$.

- **Definition clauses** (**h**, true QBFs):

$$\mathbf{h} \ id \ l_1 \ l_2 \ 0$$

storing the sequential clause identifier id and the binary definition clause $(t_i \vee \neg \ell)$ defining the auxiliary variable t_i .

- **Reference clauses** (**n**, true QBFs):

$$\mathbf{n} \ id \ l_1 \dots l_k \ 0 \ o_1 \dots o_k \ 0 \ a_1 \dots a_m \ 0$$

containing three zero-terminated sequences:

1. The literals $l_1 \dots l_k$ forming the reference clause R^σ ;
2. The inline literal origins $o_1 \dots o_k$, supplying the 1-based index of the originating matrix clause $D_j \in \phi$ for each corresponding literal l_j ;
3. The inline candidate existential witness assignment $a_1 \dots a_m$, recording the explicit existential assignment $\sigma : E_\Phi \rightarrow \{\perp, \top\}$ explored on this branch.

Refutation Trace. The propositional refutation is recorded as a sequence of derived resolution clauses:

$$\mathbf{d} \ id \ l_1 \dots l_k \ 0 \ c_1 \dots c_p \ 0$$

where id is the derived clause identifier, $l_1 \dots l_k$ are the resolvent literals (empty for the final refuting clause $\perp$), and $c_1 \dots c_p$ are antecedent clause identifiers in the proof.

2.4 Certificate Verification

DualCheck verifies certificates via a two-phase architecture:

1. **Phase 1 (Expansion Check):** The checker first processes the **x** lines to recover the mapping from each propositional variable to its original QBF variable and the assignment under which it was instantiated. Using these mappings, the checker validates that every initial clause in the certificate is a sound $\forall$ -Exp+Res axiom of Φ (for false QBFs) or of $\text{PG}(\neg\Phi)$ (for true QBFs).
2. **Phase 2 (Refutation Check):** Validates that the derived resolution steps (**d** lines) form a sound refutation deriving the empty clause $\perp$.

False QBF Verification ($s = 0$). For false instances, Phase 1 validates each axiom clause (**a** line) against its originating matrix clause $D = \phi[\text{orig}]$ by executing the expansion check defined in Section 1.5: verifying existential correspondence under ε and ensuring that the accumulated literal annotations consistently falsify all universal literals in D_U .

True QBF Verification ($s = 1$). For true instances, Phase 1 validates the initial clauses of the certificate against the matrix ϕ and the candidate existential assignments σ recorded on each reference clause (**n** line):

1. *Cube encoding and definition validation:* For each auxiliary literal $\neg t_i$ in a reference clause with originating matrix clause $D = \phi[o_j]$, the checker validates that the definition clauses (**h** lines) for t_i are of the form $(t_i \vee \neg \ell)$ such that the literals $\{\ell\}$ constitute precisely the universal subclause D_U .
2. *Annotation agreement:* The explicit existential assignment σ recorded in the reference clause agrees with the assignment recovered from the annotation of each literal $\ell \in D_U$.
3. *Clause satisfaction:* Every matrix clause omitted from the reference clause (claimed to be eliminated on this branch) is verified to be satisfied by the assignment ($\sigma \models C_E$).
4. *Definition liveness:* Every auxiliary variable defined in a definition clause (**h** line) appears negated in at least one reference clause, ensuring that no unreferenced definitions remain in the certificate.

Both checking routines operate in linear time with respect to the certificate and matrix size:

Theorem 2 (Complexity of Expansion Checking). *Phase 1 executes in time $O(|\mathcal{C}| \cdot \|\phi\|)$, where $|\mathcal{C}|$ is the number of clauses in the certificate and $\|\phi\|$ is the size of the input QBF matrix.*

Proof. For false QBFs, Phase 1 validates each of the $O(|\mathcal{C}|)$ axiom clauses in $O(|D|) \leq O(\|\phi\|)$ time by verifying literal correspondence and universal falsification via direct array lookups. For true QBFs, validating each reference clause involves checking definition clauses and literal annotations in $O(|D_U|) \leq O(\|\phi\|)$

time. Verifying clause satisfaction for omitted matrix clauses requires testing $\sigma \models C_E$, which takes $O(\|\phi\|)$ total time per reference clause. Definition liveness is verified in a single linear pass over the certificate clauses in $O(|\mathcal{C}|)$ time. Summing over all certificate clauses yields an overall verification time of $O(|\mathcal{C}| \cdot \|\phi\|)$. $\square$

3 Skolem Strategy Extraction

Beyond deciding truth values and verifying proofs, a primary practical application of QBF solving is strategy synthesis: extracting winning moves for the winning player as computable Boolean circuits. For false QBFs, strategy extraction from $\forall$ -Exp+Res refutations is well understood: symmetric Craig interpolation produces an admissible winning Herbrand strategy [7, 9]. For true QBFs, however, strategy extraction in expansion solving has remained an open challenge due to the clausal disruption introduced by structure-preserving definitional transformations. We now show that the clausal duality established in Section 2 resolves this challenge, enabling polynomial-time extraction of winning Skolem strategies from dual expansion refutations.

Eliminating Auxiliary Variables via Definition Substitution. Now let $\Phi = \Pi.\phi$ be a true QBF, and let π be the resolution refutation of the dual propositional instance produced by an expansion solver. By Theorem 1, π is an exact $\forall$ -Exp+Res refutation of the encoded negation $\text{PG}(\neg\Phi)$. In $\text{PG}(\neg\Phi)$, the quantifier roles are inverted relative to Φ : the original existential variables E_Φ become universal ($U_{\text{PG}(\neg\Phi)} = E_\Phi$), while the original universal variables U_Φ and the auxiliary definition variables T become existential ($E_{\text{PG}(\neg\Phi)} = U_\Phi \cup T$).

By the strategy extraction framework of Slivovsky [9], symmetric Craig interpolation applied to an $\forall$ -Exp+Res resolution refutation DAG directly yields an admissible winning Herbrand strategy. Applying this extraction to π on $\text{PG}(\neg\Phi)$ yields an admissible winning Herbrand strategy $h = (h_e)_{e \in E_\Phi}$ for $\text{PG}(\neg\Phi)$, guaranteeing that $\phi'[h] \equiv \perp$. However, because π contains resolution steps on auxiliary definition variables $t_i \in T$ that precede some $e \in E_\Phi$ in the prefix Π' ($t_i <_{\Pi'} e$), the component circuits h_e may read variables in T . Because T are auxiliary variables not present in Φ , h cannot serve directly as a Skolem strategy for Φ . To eliminate the auxiliary variables, we substitute each t_i with its defining disjunction:

$$\psi^* = \psi[t_1 / \bigvee_{\ell \in C_{1,U}} \ell] \dots [t_m / \bigvee_{\ell \in C_{m,U}} \ell]. \quad (20)$$

Applying this substitution to each component circuit h_e yields the substituted strategy $h^* = (h_e^*)_{e \in E_\Phi}$.

Lemma 2 (Strategy Correspondence). *If h is a winning Herbrand strategy for $\text{PG}(\neg\Phi)$, then h^* is an admissible winning Herbrand strategy for $\neg\Phi$.*

Proof. Admissibility. Let $e \in E_\Phi = U_{\neg\Phi}$. Any variable $w \in \text{vars}(h_e^*)$ either appears directly in h_e (in which case $w <_{\Pi'} e$ by the admissibility of h), or was introduced by substituting for an auxiliary variable $t_i \in \text{vars}(h_e)$ with $t_i <_{\Pi'} e$. In the latter

case, $w \in \text{vars}(C_{i,U})$. By construction of the prefix Π' (Definition 2), t_i is placed immediately after $v_i^* = \max_{<\Pi} \text{vars}(C_{i,U})$. Therefore, $w \leq_{\Pi} v_i^* <_{\Pi'} t_i <_{\Pi'} e$. Since $w \in U_{\Phi} = E_{\neg\Phi}$ and $e \in E_{\Phi} = U_{\neg\Phi}$, their relative order is preserved under quantifier inversion: $w <_{\Pi} e$. Hence $\text{vars}(h_e^*) \subseteq D_{\Pi}(e)$, confirming that h_e^* depends exclusively on variables strictly preceding e in the original prefix Π .

Soundness. To prove that h^* is a winning Herbrand strategy for $\neg\Phi$, we must show that $(\neg\phi)[h^*] \equiv \perp$. Recall from Definition 2 that the matrix ϕ' of $\text{PG}(\neg\Phi)$ is the conjunction of the definition clauses and the reference formula $\bigvee_{C_i \in \phi} (\neg C_{i,E} \wedge \neg t_i)$. Because h is a winning Herbrand strategy for $\text{PG}(\neg\Phi)$, it falsifies the matrix: $\phi'[h] \equiv \perp$. We evaluate the effect of the definition substitution $(\cdot)^*$ in four steps:

1. *Definition clauses star to $\top$:* For each clause $C_i \in \phi$ and each literal $\ell \in C_{i,U}$, applying $[t_i / \bigvee_{\ell' \in C_{i,U}} \ell']$ gives:

$$(t_i \vee \neg\ell)^* = \left(\bigvee_{\ell' \in C_{i,U}} \ell' \right) \vee \neg\ell \equiv \top,$$

due to the complementary literal pair $\ell \vee \neg\ell$. Hence all definition clauses evaluate to $\top$ under $(\cdot)^*$.

2. *The reference formula stars to $\neg\phi$:* Applying $(\cdot)^*$ to the reference formula replaces each $\neg t_i$ with $\neg(\bigvee_{\ell \in C_{i,U}} \ell) = \bigwedge_{\ell \in C_{i,U}} \neg\ell = \neg C_{i,U}$. Because $C_{i,E}$ contains no auxiliary variables from T , we have:

$$\begin{aligned} \left(\bigvee_{C_i \in \phi} (\neg C_{i,E} \wedge \neg t_i) \right)^* &= \bigvee_{C_i \in \phi} (\neg C_{i,E} \wedge \neg C_{i,U}) \\ &= \bigvee_{C_i \in \phi} \neg(C_{i,E} \vee C_{i,U}) = \bigvee_{C_i \in \phi} \neg C_i \equiv \neg\phi. \end{aligned}$$

Combined with step 1, the entire matrix stars to $\neg\phi$: $(\phi')^* \equiv \top \wedge \neg\phi \equiv \neg\phi$.

3. *$(\cdot)^*$ and $[h]$ commute:* Because the domain of $(\cdot)^*$ is T , the domain of h is E_{Φ} , $T \cap E_{\Phi} = \emptyset$, and the substituted strategy is defined componentwise by $h_e^* = (h_e)^*$, the substitutions commute:

$$((\phi')^*)[h^*] \equiv ((\phi')[h])^*.$$

4. *Substitution preserves unsatisfiability:* Because h is winning for $\text{PG}(\neg\Phi)$, $\phi'[h] \equiv \perp$. Any substitution instance of a contradiction is contradictory:

$$((\phi')[h])^* \equiv \perp^* \equiv \perp.$$

Chaining these equivalences establishes:

$$(\neg\phi)[h^*] \equiv ((\phi')^*)[h^*] \equiv ((\phi')[h])^* \equiv \perp.$$

Therefore, h^* is an admissible winning Herbrand strategy for $\neg\Phi$. $\square$

Theorem 3 (Skolem Strategy Extraction). *Let Φ be a true PCNF QBF and let π be an $\forall$ -Exp+Res refutation of $\text{PG}(\neg\Phi)$. Applying the interpolation framework of Slivovsky [9] to π followed by the definition substitution of Lemma 2 yields an admissible winning Skolem strategy $s = (s_e)_{e \in E_\Phi}$ for Φ in time $O(|\pi| \cdot \|\phi\|)$.*

Proof. We construct the winning Skolem strategy s in three stages:

1. *Refutation Structure:* By Theorem 1, π is a valid $\forall$ -Exp+Res refutation DAG of $\text{PG}(\neg\Phi)$ deriving the empty clause $\perp$. The leaves of π are universal expansion axioms of $\text{PG}(\neg\Phi)$ under candidate universal assignments $\sigma : U_{\text{PG}(\neg\Phi)} \rightarrow \{\perp, \top\}$ (where $U_{\text{PG}(\neg\Phi)} = E_\Phi$). Each internal node $v \in \pi$ represents a propositional resolution step on an annotated existential pivot literal x^τ , where $x \in E_{\text{PG}(\neg\Phi)} = U_\Phi \cup T$.
2. *Interpolation and Herbrand Strategy Synthesis:* Following Slivovsky [9], symmetric Craig interpolation applied to the refutation DAG π extracts an admissible winning Herbrand strategy $h = (h_e)_{e \in E_\Phi}$ for $\text{PG}(\neg\Phi)$ in time $O(|\pi|)$, ensuring that $\phi'[h] \equiv \perp$.
3. *Definition Substitution and Strategy Duality:* Because π contains resolution steps on auxiliary variables $t_i \in T$ with $t_i <_{\Pi'} e$, each circuit h_e may read inputs from T . Applying the definition substitution of Eq. (20) replaces each auxiliary input t_i with the disjunction $\bigvee_{\ell \in C_{i,U}} \ell$. Since each clause $C_i \in \phi$ has $|C_{i,U}| \leq \|\phi\|$, this substitution adds at most $O(|T| \cdot \|\phi\|)$ gates and executes in linear time $O(|h| + |T| \cdot \|\phi\|) = O(|\pi| \cdot \|\phi\|)$. By Lemma 2, the substituted circuit family $h^* = (h_e^*)_{e \in E_\Phi}$ is an admissible winning Herbrand strategy for $\neg\Phi$, satisfying $(\neg\phi)[h^*] \equiv \perp$. By strategy duality under negation (Remark 1), setting $s = h^*$ yields an admissible winning Skolem strategy for Φ satisfying $\phi[s] \equiv \top$.

The entire extraction procedure operates in time $O(|\pi| \cdot \|\phi\|)$. $\square$

4 Related Work

5 Conclusion

References

1. Beyersdorff, O., Blinkhorn, J., Mahajan, M.: Building strategies into QBF proofs. *Journal of Automated Reasoning* **65**(1), 125–154 (2020). <https://doi.org/10.1007/s10817-020-09560-1>
2. Bloem, R., Braud-Santoni, N., Hadžić, V., Egly, U., Lonsing, F., Seidl, M.: Expansion-based QBF solving without recursion. In: 2018 Formal Methods in Computer Aided Design (FMCAD). pp. 1–10. IEEE (2018). <https://doi.org/10.23919/FMCAD.2018.8603004>
3. Hadžić, V., Bloem, R., Shukla, A., Seidl, M.: FERPMODELS: A certification framework for expansion-based QBF solving. In: 2022 24th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNAS). pp. 80–83. IEEE (2022). <https://doi.org/10.1109/SYNASC57785.2022.00022>

Table 1. Summary of foundational symbols, notations, and domains.

Symbol	Type	Domain / Definition	Role / Meaning
<i>Formula Syntax and Semantics</i>			
$\Phi = \Pi.\phi$	PCNF QBF	$\phi = \{C_1, \dots, C_m\}$	Input formula with linear prefix Π and CNF matrix ϕ
E_Φ, U_Φ	Variable sets	$V_\Phi = E_\Phi \cup U_\Phi$	Existential and universal variable sets of Φ
$D_\Pi(v)$	Variable set	$\{w \in V_\Phi \mid w <_\Pi v\}$	Dependency domain of variable v under Π
$C_i = C_{i,E} \vee C_{i,U}$	Clause	$\text{vars}(C_{i,E}) \subseteq E_\Phi, \text{vars}(C_{i,U}) \subseteq U_\Phi$	Clause partitioned into existential and universal parts
$\neg\Phi = \overline{\Pi}.\neg\phi$	PDNF QBF	$U_{\neg\Phi} = E_\Phi, E_{\neg\Phi} = U_\Phi$	Negation of Φ ; false QBF targeted in dual solving
<i>Strategies and Duality</i>			
$s = (s_e)_{e \in E_\Phi}$	Skolem strategy	$\text{vars}(s_e) \subseteq D_\Pi(e)$	Winning strategy for Φ ($\phi[s] \equiv \top$)
$h = (h_u)_{u \in U_\Phi}$	Herbrand strategy	$\text{vars}(h_u) \subseteq D_\Pi(u)$	Winning strategy for false Φ ($\phi[h] \equiv \perp$)
<i>Expansion Solving and Proof Calculus</i>			
α	Assignment	$\alpha : U_\Phi \rightarrow \{\perp, \top\}$	Universal assignment explored in primal solving
σ	Assignment	$\sigma : E_\Phi \rightarrow \{\perp, \top\}$	Existential candidate assignment explored in dual solving
ϕ^α	Prop. CNF	$\bigwedge_{C \in \phi} C^\alpha$	Instantiation of ϕ under universal branch α
ϕ^σ	Clause set	$\{C_i \in \phi \mid \sigma \not\models C_{i,E}\}$	Clauses unsatisfied by existential candidate σ
$\neg\phi^\sigma$	Prop. DNF	$\bigvee_{C_i \in \phi^\sigma} \neg C_{i,U}^\sigma$	Negated instantiation on branch σ
$\forall\text{-Exp+Res}$	Proof system	Universal expansion + resolution	Background proof calculus for expansion refutations
<i>Clausal Negation Encoding and Strategy Extraction</i>			
$t_i \in T$	Aux. variable	$v_i^* <_{\Pi'} t_i$	Abbreviation for universal subclause $\bigvee_{\ell \in C_{i,U}} \ell$
v_i^*	Variable	$\max_{<_\Pi} \text{vars}(C_{i,U})$	Innermost universal variable of C_i in Π
$\text{PG}(\neg\Phi) = \Pi'.\phi'$	QBF	$U_{\text{PG}(\neg\Phi)} = E_\Phi, E_{\text{PG}(\neg\Phi)} = U_\Phi \cup T$	Definitional encoding of $\neg\Phi$
$(t_i \vee \neg\ell)$	Clause	$\ell \in C_{i,U}, t_i \in T$	Definition clause encoding $\neg t_i \rightarrow \bigwedge_{\ell \in C_{i,U}} \neg\ell$
R^σ	Clause	$\bigvee_{C_i \in \phi^\sigma} \neg t_i^\sigma$	Reference clause emitted on branch σ
ψ^*	Substitution	$[t_i / \bigvee_{\ell \in C_{i,U}} \ell]$	Definition substitution eliminating auxiliary variables
I_i^C	Boolean circuit	Formula over $D_\Pi(u_i)$	Partial Craig interpolant label for clause C and variable u_i
$G_i(\mu)$	Guard condition	$H_\mu^{i-1} \wedge (\mu(u_i) \leftrightarrow 0)$	Prefix guard matching opposing assignment μ

4. Janota, M., Klieber, W., Marques-Silva, J., Clarke, E.M.: Solving QBF with counterexample guided refinement. *Artificial Intelligence* **234**, 1–25 (2016). <https://doi.org/10.1016/j.artint.2016.01.004>
5. Janota, M., Marques-Silva, J.: Expansion-based QBF solving versus Q-resolution. *Theoretical Computer Science* **577**, 25–42 (2015). <https://doi.org/10.1016/j.tcs.2015.01.048>
6. Plaisted, D.A., Greenbaum, S.: A structure-preserving clause form translation. *Journal of Symbolic Computation* **2**(3), 293–304 (1986). [https://doi.org/10.1016/S0747-7171\(86\)80028-1](https://doi.org/10.1016/S0747-7171(86)80028-1)
7. Schlaipfer, M., Slivovsky, F., Weissenbacher, G., Zuleger, F.: Multi-linear strategy extraction for QBF expansion proofs via local soundness. In: *Theory and Applications of Satisfiability Testing – SAT 2020*. *Lecture Notes in Computer Science*, vol. 12178, pp. 429–446. Springer (2020). https://doi.org/10.1007/978-3-030-51825-7_30
8. Simader, M., Rebola-Pardo, A., Seidl, M.: FERAT: A new expansion-based certification framework for quantified boolean formulas. In: *Proceedings of the 40th ACM/SIGAPP Symposium on Applied Computing (SAC)*. pp. 1043–1050. ACM (2025). <https://doi.org/10.1145/3672608.3707863>
9. Slivovsky, F.: Strategy extraction by interpolation. In: Chakraborty, S., Jiang, J.H.R. (eds.) *27th International Conference on Theory and Applications of Satisfiability Testing (SAT 2024)*. *Leibniz International Proceedings in Informatics (LIPIcs)*, vol. 305, pp. 28:1–28:20. Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2024). <https://doi.org/10.4230/LIPIcs.SAT.2024.28>